

Lahiru Wijesundara, Luise Rivaldo Lim and Omer Cem Ozdemir

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Assessment – Dissertation Paper

Driving Down Emissions: A Serious Game Approach to Transport Decarbonisation

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1 INTRODUCTION

The United Kingdom has taken efforts to reduce carbon emissions, in accordance with the Paris Agreement, to reverse the effects of global warming and ensure that global temperature does not rise by 1.5°C (Gudde et al, 2021). Although the UK is one of the first nations to participate in the agreement, results have varied between relative success (Collings, 2023) and unsatisfactory results (Gudde et al, 2021) despite having clear plans for decarbonisation policies. (Kinsella et al, 2023, Purwienanti, & Purwanto, 2023). It is suggested that the public's lack of awareness of decarbonisation policies and socio-political rejection of environmental values contribute to disappointing decarbonisation attempts (Perlaviciute, Steg & Sovacool, 2021).

Serious games and simulators have been used for the purposes of research and education (Barcena-Vazquez, Caro, Zatarain, 2023), reporting varying results across different fields of study (Waddington & Fennewald, T, 2018). It can be used for both education and recreation to great effectiveness (Fernández Galeote et al, 2021). However, the creation and implementation of serious games in the education sector is complex, requiring careful planning and research on methodologies, target demographics and field of study to perform effectively (Barcena-Vazquez, Caro, Zatarain, 2023; de Salas et al, 2022). Despite the difficulty of designing a serious game, they have previously yielded positive results based on studies conducted by and from the use of them (Fernández Galeote et al, 2021).

To find the effectiveness of serious games in education, the topic of climate change and net-zero emission plans was selected due to the gap in public knowledge. This paper follows the development of a serious game intended to educate players about the urgency of achieving net-zero carbon emissions by the year 2050, following the Paris Agreement. Iterative playtesting with participants is conducted, with each session ending with a questionnaire and feedback session discussing the educational and recreational factors of the game. By identifying the key flaws and strengths of the game components based on their educational and recreational impact, they are iteratively modified and improved upon, with the subsequent playtesting session used to gather more information.

The project aims to have participants play the role of a policymaker to implement socio-political policies to achieve sustainability in the private and public sectors. Different game elements will correspond to the players' actions and are visually represented based on the simulation data.

Through iterative playtesting and feedback collection, the serious game's effectiveness in educating carbon decarbonisation strategies is assessed, while different game elements are identified as affecting learning effectiveness.

1.1 Research Questions:

How effective is a serious game as an informative tool?

Past studies have tried to measure the effectiveness of serious games used in education. However, many papers have acknowledged flaws in their research and methodologies, accounting for variables such as sample group and field of study. This project aims to contribute to research conducted by existing papers by implementing our own unique methodologies of playtesting and surveying to identify the strengths and weaknesses of serious games

What game elements in a serious game affect the player experience, and how do they correlate to learning objectives?

By designing our own serious game, we have much better control over elements and variables that could affect the research result. By conducting frequent iterative testing on a number of participants, this project aims to identify which game elements are most effective in promoting player immersion, motivation, and knowledge retention, correlated to the serious game's intended learning objectives.

How to conduct research using serious games to find consistent and reliable research data?

There are many variables affecting a participant's engagement with a game, and the complexity of research questions can cause difficulties in observing player behaviour, correlating experiences to research data, and drawing accurate conclusions based on surveys and questionnaires. This project will aim to establish best practices for

conducting research using serious games to generate valid and reliable data by using research methodologies and customised questionnaires and surveys linked to the specific learning objectives.

2 LITERATURE REVIEW

The United Kingdom is the first country to act upon the goal of reaching net-zero carbon emissions by transitioning from fossil fuels to low-carbon alternatives (Gudde et al, 2021). It was planned to reduce carbon emissions to 35% of the current average emissions (from the early 2000s) by 2035, with further reduction by 2050 (Collings, 2023). Several studies have shown evidence that carbon emissions have since reduced (Collings, 2023). However, there have been resistances and challenges to decarbonisation attempts within the public sector, citing issues linked to economic growth, transport, housing and leisure (Gudde et al, 2021). Perlaviciute, Steg and Sovacool (2021) claim that people are generally hesitant to the transition to clean energy due to social influences such as changes to lifestyle, technological acceptance, and perception of net-zero energy systems. The negative public perception of clean energies greatly affects decarbonisation policies as it is proven that the transition to clean energy, particularly in the transport sector (Kinsella et al, 2023), is a major contributor to the reduction of CO₂ emissions. As the UK is falling behind expected goals and projections of carbon goals compared to other nations (Purwienanti & Purwanto, 2023), there is an urgency to improve awareness of decarbonisation within the public sector.

The use of serious games in education and supporting awareness has been criticised in the past (de Salas et al, 2022, Rooney-Varga et al, 2018, Waddington & Fennewald, 2018), with many papers utilising serious games in research but lacking the proper reasoning or justification to use them (de Salas et al, 2022). There is evidence to support that serious games can improve engagement in learning and improve knowledge and awareness of targeted subjects of participants (Barcena-Vazquez et al, 2023; Fernández Galeote et al, 2021). Participants who were involved in studies measuring the effectiveness of serious games in promoting engagement and learning generally reported positively to the medium used (Barcena-Vazquez et al, 2023, Fernández Galeote et al, 2021) despite flaws that may be present when engaging with serious games as a tool (Waddington & Fennewald, 2018).

To understand how to effectively design and use serious games as a learning tool, several analyses and design studies have been conducted to establish best practices in the development and execution of a serious game. Fernández Galeote et al (2021) identified different variables that affect the effectiveness of serious games used in data gathering, suggesting that the medium of delivery is an important factor when designing a game, stating that digital experiences resulted in less engagement compared to hybrid and analogue games due to the lack of social elements of the former. The target audience of conducted studies should also be appropriate to the research objectives (Rooney-Varga et al, 2018, Waddington & Fennewald, 2018), as prior knowledge to the subject, personal biases and history, and motivation levels all affect the effectiveness of serious games on an individual level (Rooney-Varga et al, 2018), causing difficulties when designing standardised tests across multiple disciplines.

Developers must be wary of various contexts and information prior to designing and developing a serious game. Fernández Galeote et al (2021) recommend implementing game-based experiences that tie into the learning objectives by focusing on specific behaviours and impacts of the research subject. Rooney-Varga et al (2018) suggest that humans can only comprehend dynamic interactions of two to three variables at a time and suggest constraints in simulations to be scientifically grounded and capable of engaging based on the capabilities of participants. The use of psychological techniques in engagement and data collection is also a factor to be considered, as different behavioural techniques can affect the quantity and quality of data collected. In a comprehensive literature study conducted by de Salas et al (2022), they found that "reward and threat mechanics" is the most used technique in serious games, with games that used "behaviour comparisons" being the least adopted technique. The heavy use of design technologies should also be considered as design visuals, and providing positive feedback upon completing an action is important to increase willingness in engagement and motivation (Barcena-Vazquez et al, 2023; Rooney-Varga et al, 2018).

Studies conducted on serious games regarding climate change are not new. Rooney-Varga et al (2018) conducted a study using a role-playing game, "World Climate", where participants are tasked to act as UN climate negotiators. Rooney-Varga et al show that participants showed an increase in knowledge of climate change after playing the game, with increased levels of motivation to act upon the knowledge gained. Rooney-Varga et al also found that the use of urgency

was ineffective against participants who lacked hope in the subject. Barcena-Vazquez et al (2023) developed a game to support global warming awareness named "Reto Global" and had players engage with the game and answer surveys after playing. While engagement was positive with reported increased awareness of the topic after playing, they acknowledge flaws in the study due to the minimal sample size and demographic bias. Waddington & Fennewald (2018) studied the game "Fate of the World (2011)" to discuss how the game impacts players' perception of climate change. Their research shows that while participants gained knowledge of global warming, engagement was unaffected. They argue that this was due to the differing ideologies of players and flaws that impacted Fate's effectiveness as a game and an educational tool.

3 BACKGROUND

The importance of addressing climate change and reducing carbon emissions has increased in urgency as we are reaching the first deadline of reducing carbon emissions to 30% by the year 2035 (Collings, 2023). Despite efforts made by the UK to uphold those promises, the results have been mixed, with varying degrees of success reported in different studies (Gudde et al, 2021; Collings, 2023). Challenges such as the lack of public awareness and socio-political resistance to environmental policies have hindered progress towards achieving net-zero emissions by 2050 (Kinsella et al, 2023; Perlaviciute, Steg & Sovacool, 2021).

It is acknowledged that serious games have emerged as promising tools for education and building public awareness on scientific topics such as climate change and sustainability (Barcena-Vazquez, Caro, Zatarain, 2023). However, existing research on serious games has highlighted the need for more robust methodologies and targeted approaches to maximise their effectiveness (de Salas et al., 2022; Rooney-Varga et al, 2018; Waddington & Fennewald, 2018). Despite their potential, serious games have faced criticism for their implementation and lack of justification in educational contexts (de Salas et al, 2022).

This research project seeks to address these gaps in the literature by developing and evaluating a serious game focused on carbon decarbonisation and net-zero emission plans in the transport sector. By using the engaging nature of serious games, the aim is to improve public awareness and understanding of decarbonisation strategies and policies by utilising recreational elements found in video games to motivate learning objectives. Additionally, the project aims to contribute

to the design of serious games by analysing the impact of different game elements on player experience and learning outcomes.

By conducting iterative playtesting sessions and collecting feedback from participants, this research project aims to refine the serious game iteratively, ensuring that it effectively communicates key concepts related to carbon decarbonisation while providing an engaging and enjoyable gameplay experience. Through these methods, the project seeks to contribute to the discussion of the impact of serious games in addressing societal challenges and promoting sustainability education.

4 METHODOLOGY

4.1 Game Planning and Design

In order to incentivise players to prioritise issues regarding the sustainability of transport vehicles and decarbonisation policies, the focus of the game should relate to transport vehicles and the harmful effects of their carbon emissions. Research was conducted regarding the carbon emissions emitted by various vehicle types in the area of Bristol, Great Britain. The following table shows carbon emissions emitted by different vehicles based on their distance travelled during that year:

Vehicles	kg Co2/mile	Miles travelled per year	Initial % of mode use	Emissions in KG per year
Train	0.02	720	4	14.4
Car	0.291	5086	51	1480.026
Bus	0.044	385	15	16.94
Bicycle	0	57	10	0
Walking	0	87	20	0

Table 1. Vehicle Emissions Data from Bristol, Great Britain

Attention was drawn to several commercial games that focus on the use of vehicular transport in economic and environmental contexts. In the game *Cities Skylines* (2015), players take the role of a city planner focused on the development and expansion of an urban city. Careful placement of buildings that house inhabitants and economic buildings that generate income tie into the construction of roads where vehicles travel between locations for virtual inhabitants to conduct commercial activities. The game encourages balancing of different social, economic and environmental factors in order to run an economically viable city while making sure that social and environmental requirements are met. As the city expands, the virtual inhabitants expand in number, resulting in higher demands for income production and the construction of roads and highways to allow the use of transport vehicles for commuting between locations. By focusing on economic growth, players run the risk of ignoring environmental factors as the use of transport increases pollution and toxicity in the air, negatively affecting the welfare of the inhabitants. *Mini Motorways* (2015) is a video game where players construct roads and bridges to connect different buildings together, allowing travel between them. Players must ensure that all buildings are connected to their target destinations by building roads, while vehicles are subject to traffic conditions, failing the game if vehicles are unable to reach their destinations within a time limit. As the volume of vehicles increases and land space is finite, players will struggle designing road systems that accommodate the large number of vehicles travelling within the city. Both games emphasise careful city planning and accommodate a growing population of vehicles, highlighting the struggles of maintaining a large number of vehicles within a community. This aligns with the research objective of maintaining energy sustainability as the usage of personal vehicles increases.

As a policymaker, players are able to observe statistics and directly manipulate game data to cause an impact on the virtual environment. Players are provided a set of policies which affect the usage and distribution of vehicles. In order to make the game winnable, electric vehicles that emit no carbon emissions were added to the game, allowing players to reach the goal of net-zero emissions through the manipulation of vehicle usage.

ID	Name	Description	Initial amount	Tokens per Unit	Implementation Time per Unit (Months)	Cost (Tokens)	Policy Effect
1	Replace Gas Car with Gas Bus	Encourage car drivers to use more buses.	424	30	0.35	1500	Replace the amount of Gas Cars with Gas Buses
2	Replace Gas Car with Gas Train	Encourage car drivers to use more trains.	424	36	0.42	1800	Replace the amount of Gas Cars with Gas Trains
3	Build cycling lane	Encourage citizens to cycle more.	7.25	100	0.7	5000	Replace the amount of Gas Cars with Bicycles
4	Build walking lane	Encourage citizens to walk more.	4.75	120	0.85	6000	Replace the amount of Gas Cars with Pedestrians
5	Upgrade Gas Car to Electric Car	Upgrades gas cars to electric cars.	424	55	0.24	2750	Replace the amount of Gas Cars with EV Cars
6	Upgrade Gas Bus to Electric Bus	Upgrades gas buses to electric buses.	32	50	0.24	2500	Replace the amount of Gas Bus with EV Buses
7	Upgrade Gas Train to Electric Train	Upgrades gas trains to electric trains.	60	65	0.3	3250	Replace the amount of Gas Trains with EV Trains
8	Plant Trees	Plants trees around the city.	0	3000	12	3000	Plant Trees around the city

Table 2. Implementable Policies Table

Carbon emissions for each vehicle are calculated based on its mileage, which reflects its frequency of use and circulation. This calculation involves multiplying the vehicle's mileage by the CO2 emissions released per mile, as specified in the vehicle emissions table.

$$\text{Vehicle Emissions} = \text{Mileage} * \text{Emissions}$$

Since each vehicle has its own unique mileage and emission values, these factors are tracked independently. To standardise the calculation, values are adjusted to be calculated in in-game months, which consist of 30 in-game days. The game also maintains a cumulative carbon emissions total, tracking the total emissions released each year from all active vehicles. This cumulative footprint is visually represented within the game interface, allowing players to observe the environmental impact of their choices over time.

The game starts from the year 2021. To provide a sense of urgency for the player, the game has milestones set for the years 2030 and 2050. By 2030, the player must have reduced initial carbon emissions from 2021 by 45% and have reached net-zero emissions by 2050. If either condition has failed to be met, the game will be over, and the player will have lost the game. Additionally, to simulate the warning of exceeding the global warming threshold of 1.5°C, a loss condition is added where the game will automatically end if accumulated carbon emissions have exceeded 20,000 KG (20 tonnes).

4.2 Implementation and Technical Documentation

4.2.1 Game engine

The game uses Unity3d engine for its functionality on web browsers. It provides a virtual environment to test the impact of different transport modes and policies on carbon emissions. The game features realistic transport modes, including gas and electric cars, buses, trains, bicycles, and pedestrian options.

4.2.2 Source Control

GitLab was used for source control to ensure systematic version control and collaboration management. The game's assets, including materials, scripts, and images, were organised based on predefined naming conventions to facilitate ease of modifications and ensure a structured development process.

4.2.3 Game Mechanics

- 4.2.3.1 ***Statistics Tracking and Retention:*** *The game accurately tracks various metrics such as the distance travelled by each mode of transport, monthly and total carbon emissions, and the effectiveness of the implemented policies over a simulated period. Players can observe the impact of their choices on carbon emissions.*
- 4.2.3.2 ***Transport Mode Simulation:*** *Different modes of transportation have varying CO2 emissions and mileage. The sum of all transport modes' emissions provides the total carbon output.*
- 4.2.3.3 ***Policy Implementation:*** *Players can apply different policies, which are processed in real time to adjust the distribution of transport modes and calculate the subsequent emissions.*

4.2.4 Development Assets

Due to time constraints, third-party assets sourced from the Unity Asset Store were used to facilitate building and vehicle assets, particle effects, and sound effects.

4.3 Development

Development of the game consists of three major milestones that marks the level of maturity of the game's overall quality and performance: prototype, alpha and beta stages.

4.3.1 Prototype Stage

The prototype phase is intended to start the foundation of the game's mechanics and provide an educational experience regarding carbon emissions reduction and sustainable urban planning. However, several key aspects remained underdeveloped or unclear during this initial phase.



Image 1. Screenshot of the game during its prototype stage

a) Policies:

- Out of the 8 planned policies, only 3 were successfully implemented:
 - Policy 1: Reduce car mileage by 20%, increase EV car mileage by 20%
 - Policy 2: Reduce bus mileage by 20%, increase EV bus mileage by 20%
 - Policy 3: Reduce train mileage by 20%, increase EV train mileage by 20%
- Due to the reliance on percentages when applying policies, it is impossible to reach the goal of net-zero emissions.

b) City Environment:

- The city design lacked any colour or detail to make it a believable urban city.
- The train track meant to visualise the movement of rail systems was incomplete and lacking texture.

c) Vehicle Movement:

- Vehicle movement within the city was not functional as cars remained stationary at points placed around the map.

- The lack of movement introduced difficulty for play testers to visualise the player's actions on the environment.
- d) **Pollution/Negative Impact:**
 - The depiction of pollution and its negative impact on the environment was unclear and was not communicated well to the player.
 - There was a lack of visual indicators or feedback mechanisms to highlight the consequences of high carbon emissions and unsustainable practices.
- e) **Game Objectives:**
 - The game lacks any tutorial or guidance, leading to confusion among players regarding their goals and objectives.
 - Without clear objectives, players struggled to engage with the gameplay and understand the purpose of their actions within the game world.
- f) **Policy Impacts:**
 - The direct impact of implemented policies on carbon emissions and sustainability outcomes was not evident to players due to the lack of environmental visuals and movement.
 - All player actions are locked upon implementing a policy, resulting in downtime where players are unable to do anything, resulting in disassociation while playing the game.
 - Players were unable to notice the effects or consequences of their policy decisions, resulting in no educational value received from playing the game.

4.3.2 Alpha Stage

After receiving feedback from the prototype phase, the project was developed further, and major improvements were made. Multiple planned mechanics have been successfully implemented, and learning objectives are more clearly defined.



Image 2. Screenshot of the game during its alpha stage

a) Policies:

- Out of the 8 planned policies, 7 have been successfully implemented. The new policies provide players with a more diverse range of options to influence carbon emissions and sustainability within the city.
- Policies now use the proper mathematical formula when calculating vehicle displacement and carbon emission reduction, allowing players to reach net zero emissions by converting carbon-emitting vehicles into transport with zero emissions.

b) Tutorial Scene:

- A tutorial scene has been implemented to provide players with an introduction to gameplay mechanics and objectives.
- However, the text-only format of the tutorial proved to be ineffective in fully conveying important information, indicating a need for more interactive and engaging tutorials.

c) City Environment:

- The city environment has been completely remade to include more buildings and roads.
- However, the desaturated visuals result in the city looking bland and lifeless.

d) Pollution Visuals:

- A new particle system was implemented to improve the visuals of the portrayal of carbon emission pollution.
 - However, the pollution still lacks realism and fails to effectively communicate the environmental impact of carbon emissions.
- e) Token Feedback:
- Whenever players gain tokens, an animation and sound effect will play, signalling that the player's token count has increased.
 - The added visual and audio feedback adds a layer of interactivity and rewards the player for playing by providing a pleasant experience.
- f) Vehicle Movement:
- Cars now exhibit more organic movement patterns within the city, enhancing the sense of realism and immersion.
 - However, further refinements are needed to ensure that vehicle behaviour aligns with real-world traffic dynamics.
- g) Statistics Page:
- A new statistics page has been added to provide players with detailed information on vehicles and their environmental impact.
 - This feature aims to enhance player understanding of the connections between transportation types and carbon emissions.
- h) Policy Effects:
- The effects of implemented policies are now more legible and clearly communicated to players through text descriptions and policy titles.
 - Despite this improvement, there is still a perceived lack of impact, indicating a need for stronger visual feedback to communicate the consequences of policy decisions.
- i) Help Feature:
- A help feature has been introduced to remind players of key gameplay elements and provide assistance when needed.
 - This feature serves as a valuable resource for players seeking clarification on game mechanics and objectives.

4.3.3 Beta Stage

As the game reaches its final development stages, most planned features and mechanics have been implemented, resulting in an engaging and educational gameplay experience.



Image 3. Screenshot of the game during its alpha stage

1. Policies:

- All 8 policies have been successfully implemented based on initial plans.
- The new Plant Tree policy allows players to have a method to passively reduce total carbon emissions as trees absorb CO₂.

2. New Tutorial:

- A new tutorial has been introduced, featuring forced gameplay elements that encourage players to learn by doing.
- This interactive approach to learning ensures that players understand gameplay mechanics and objectives while actively engaging with the game environment.

3. City Environment:

- The city environment has been enhanced with vibrant colours, creating a more visually appealing landscape.

4. Improved Pollution Visuals:

- Particle effects have been refined to visually depict polluted air more effectively within the game environment.

- These enhancements help players understand the consequences of carbon emissions and the urgency of implementing sustainable policies.
5. Vehicle Impact:
- Dense gas vehicles now communicate their negative environmental impact more prominently, conveying the urgency to change to cleaner transportation options.
 - This visual feedback reinforces the importance of reducing carbon emissions and encourages players to make informed decisions when selecting transportation modes.
6. Enhanced Statistics Page:
- The statistics page has been expanded to explain the impacts of implemented policies more comprehensively.
 - Additionally, a new feature has been added to track the history of implemented policies and visualize the decline in carbon emissions over time, providing players with valuable feedback on their progress.
7. Educational Tips Pop-up:
- A new tips pop-up feature has been introduced to educate players with in-game instructions and educational trivia.
 - This feature serves as a valuable resource for players seeking additional information and guidance on gameplay mechanics and sustainability concepts.

4.4 Research and Data Collection

Data collection is gathered during three major milestones of the game's development. Play testing sessions were hosted at the end of the game's prototype, alpha and beta stages. Each stage represents the maturity of game mechanics. Feedback received from the prototype stage was improved upon for the alpha playtest, and the process is repeated again for the beta playtest.

The all playtests were conducted with control groups consisting of professional adults and university students. At the end of a playtest session, participants were asked to fill a questionnaire rating their experience playing the game from a scale of 1 to 10 (1 = least impactful, 10 = most impactful). Results are compiled in a chart to be evaluated.

4.4.1 Prototype Results

Prototype	Player 1	Player 2	Player 3	Player 4
Overall Enjoyment	2	3	6	1
Engagement	4	5	6	4
Clarity of Instructions	2	3	6	1
Visuals	3	3	6	1
UI and Accessibility	3	3	4	2
Motivation to play	4	4	5	3
Understood Game Objectives	4	6	6	2
Understood Learning Objectives	6	6	8	4
Prior knowledge of subject	4	5	7	4

Table 3. Prototype playtest results

In the prototype stage, major game mechanics and visuals are still missing, so players are guided through a vocal demonstration of the game. Overall, the reception of the prototype game was negative due to its unfinished state. Many players were discouraged from continuing to play the game due to missing features and unengaging gameplay. Most players were unable to complete the game as they did not understand the game objectives and had to rely on guidance from the game makers, which impacted their ability to make independent decisions. Play testers reported that a lack of visual feedback and clear instructions strongly impacts their ability to understand the learning objectives of the game. When questioned about their familiarity with transport decarbonisation and net-zero emission policies, most responded that they were mostly unfamiliar with the topic.

4.4.2 Alpha Results

Alpha	Player 1	Player 2	Player 3	Player 4	Player 5	Player 6
Overall Enjoyment	4	4	5	6	5	4
Engagement	4	5	3	4	4	5
Clarity of Instructions	6	5	5	7	7	8
Visuals	7	6	7	5	7	6
UI and Accessibility	5	5	6	6	5	5
Motivation to play	4	4	5	6	5	4
Understood Game Objectives	6	6	5	6	6	5
Understood Learning Objectives	6	7	6	7	7	6
Prior knowledge of subject	5	5	7	4	5	4

Table 4. Alpha playtest results

Reception for the alpha game ranged from mixed to positive. The improved visuals initially improved engagement as players spent more time in the beginning navigating through the

new menus and UI to understand game mechanics. However, engagement quickly declined when they were unable to perform additional actions while implementing a policy. Few players noted that the reliance on text to deliver game tips and tutorials is not fun and demotivates them from continuing to play the game. Players noted that the moving cars with visuals of city pollutants contributed to their understanding of the game's learning objectives on transport decarbonisation. Players reported that they have learned new information regarding transport decarbonisation, although few were already familiar with the topic prior to playing the game.

4.4.3 Beta Results

Beta	Player 1	Player 2	Player 3	Player 4	Player 5	Player 6	Player 7	Player 8
Overall Enjoyment	5	5	5	6	5	5	5	4
Engagement	6	6	6	8	7	6	5	7
Clarity of Instructions	6	7	7	8	7	5	8	5
Visuals	8	8	7	9	8	7	7	8
UI and Accessibility	6	7	5	7	6	7	6	4
Motivation to play	5	6	4	4	7	6	6	4
Understood Game Objectives	8	9	7	9	8	6	9	7
Understood Learning Objectives	6	7	7	9	8	6	6	7
Prior knowledge of subject	4	4	5	7	6	5	5	7

Table 5. Beta playtest results

Reception for the beta stage was generally positive, as core gameplay mechanics have been fully implemented. Players were able to utilise the statistics pages to navigate through game elements and learn the game mechanics to reach game objectives effectively and successfully. Most players were immersed in gameplay, actively flipping through tutorials to devise strategies to complete game objectives more efficiently. The improved vehicle visuals and pollutants emitted from them incentivised players to quickly replace them with electric vehicles by implementing the related policies, which they were able to easily find using the icon-based policy table. Engagement has improved from the alpha stage, but a few players started getting demotivated after some time playing, as the repetitive nature of implementing policies and waiting was boring. Play testers reported that they have learned more about transport decarbonisation after playing the game, attributing this to the game's visual representation of vehicles and carbon pollutants and informative descriptions of policies and educational trivia. When questioned about their familiarity with transport

decarbonisation, most were ignorant or uninterested but have shown interest in learning more after playing the game.

4.4.4 Research Flaws and Weaknesses

The sample sizes of the research conducted were minimal, and we were unable to gather more diverse number of participants. Additionally, participants were gathered from the same campus, which could lead to perception bias. Most participants were not UK residents, which could have affected their familiarity and interest of the topic as most of the games policies and learning objectives are primarily centred around UK environmental policies. The survey was also self-reported, so the accuracy of the results may be doubted. Future studies would be advised to gather data from a larger and diverse sample group from different backgrounds and levels of familiarity with the topic.

5 CONCLUSION

From the results of the survey and participant feedback, several potential aspects that could influence decision-making were identified:

- Game Visuals

Many participants reported higher levels of engagement when the game's visuals were improved. This suggests that visual representation of data is crucial in a player's understanding of game concepts.

- Prior knowledge of the topic

Familiarity of the topic has little correlation to how a player engages with the game. A player must have sufficient motivation to be engaged with the game.

- Learning Objectives

There is a correlation between new information gained and the clarity of game instructions provided. When game mechanics are presented clearly and effectively, players are able to retain and gain more information.

- Game rewards after achieving milestones

The addition of token feedback in the beta stage not only improved average player engagement, but also increased their motivation to play the game. This suggests that audio-visual feedbacks contribute to making an engaging experience in a game.

- Penalties for failing milestones

Many players ignored the milestone goals until a warning pop-up was presented to them.

This suggests that players are likely to have little motivation until a sense of urgency compels them to engage more with game mechanics.

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